Activity 5.2: Trigonometric Ratios

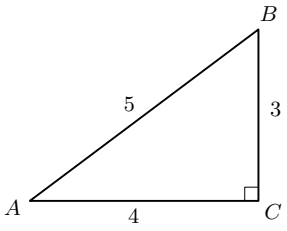
A

Name: _____ Partner's name: _____ Date: _____

Directions: Give ratios as simplified fractions. Round decimals to the nearest tenth. Use degree mode on your calculator. Diagrams are accurate, but use math rather than a ruler or protractor.

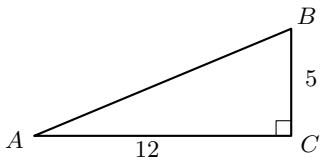
Work alone, then compare after each problem. If answers differ, find the mistake together.

1 Find $\sin A$.



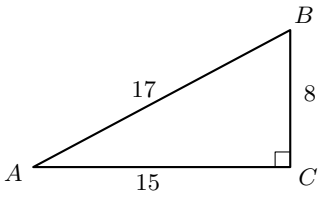
Answer: _____ Match? ☐ Yes ☐ No

2 Find $\tan A$.



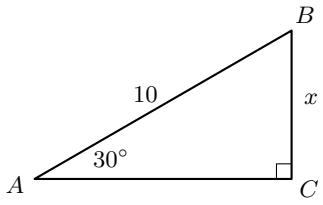
Answer: _____ Match? ☐ Yes ☐ No

3 Find $\cos B$.



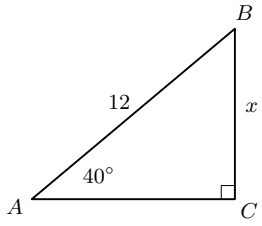
Answer: _____ Match? ☐ Yes ☐ No

4 Find x (exact).



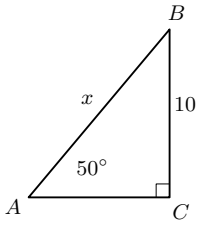
Answer: _____ Match? ☐ Yes ☐ No

5 Find x.



Answer: _____ Match? ☐ Yes ☐ No

6 Find x.



Answer: _____ Match? ☐ Yes ☐ No

7 In right $\triangle ABC$ (right angle at C), $BC = 5$ and $AC = 12$. Find $m\angle A$.

Answer: _____ Match? ☐ Yes ☐ No

8 A 20-ft ladder leans against a building and makes a $60°$ angle with the ground. How high up the building does it reach?

Answer: _____ Match? ☐ Yes ☐ No

Activity 5.2: Trigonometric Ratios

B

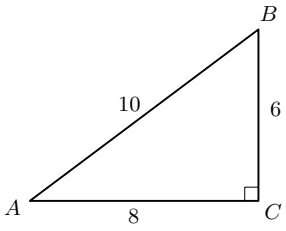
Name: _____ Partner's name: _____ Date: _____

Directions: Give ratios as simplified fractions. Round decimals to the nearest tenth. Use degree mode on your calculator. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

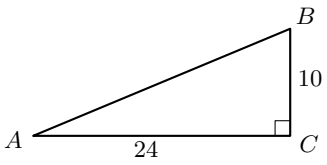
Find $\sin A$.



Answer: _____ Match? ☐ Yes ☐ No

2

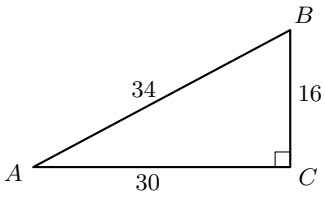
Find $\tan A$.



Answer: _____ Match? ☐ Yes ☐ No

3

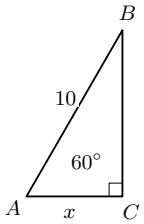
Find $\cos B$.



Answer: _____ Match? ☐ Yes ☐ No

4

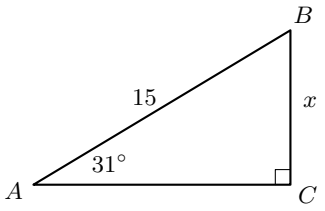
Find x (exact).



Answer: _____ Match? ☐ Yes ☐ No

5

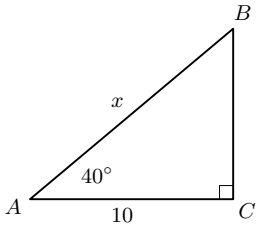
Find x.



Answer: _____ Match? ☐ Yes ☐ No

6

Find x.



Answer: _____ Match? ☐ Yes ☐ No

7

In right $\triangle ABC$ (right angle at C), $BC = 10$ and $AB = 26$. Find $m\angle A$.

Answer: _____ Match? ☐ Yes ☐ No

8

A kite string 20 m long makes a 60° angle with the ground. How high is the kite above the ground? (Assume the string is straight.)

Answer: _____ Match? ☐ Yes ☐ No